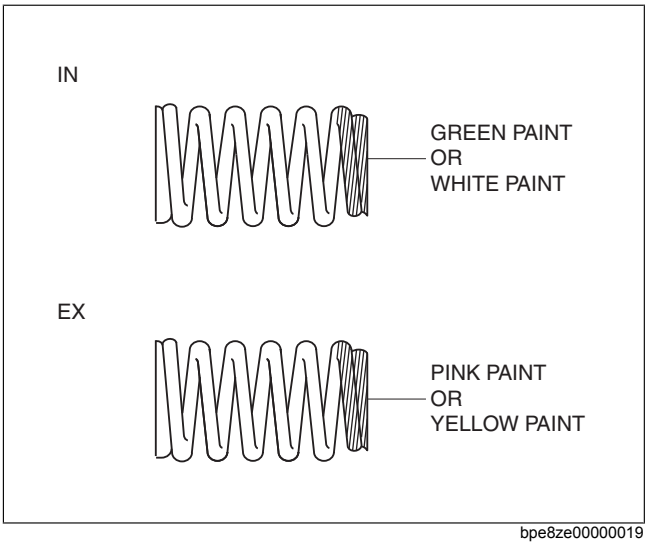
VALVE SPRING INSPECTION

id011000501400

Caution

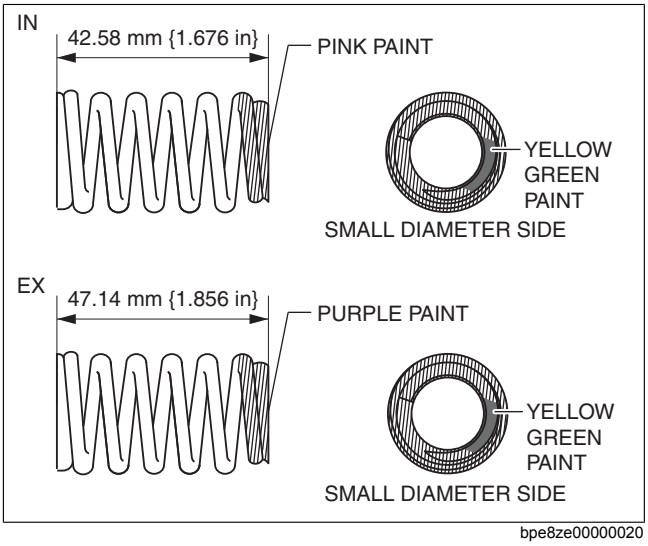
- **The valve springs differ depending on the IN and EX sides.Therefore, verify the free length or identification paint beforehand and inspect the valve springs.**

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (Except MX-5 (ND))



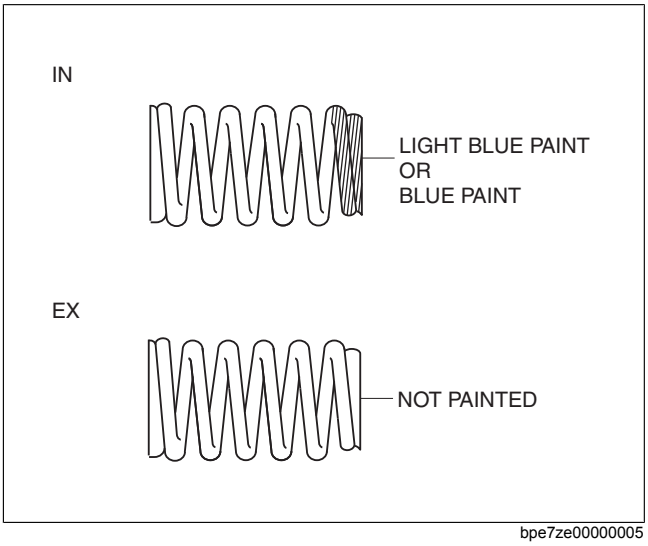
bpe8ze00000019

SKYACTIV-G 1.5 (MX-5 (ND))



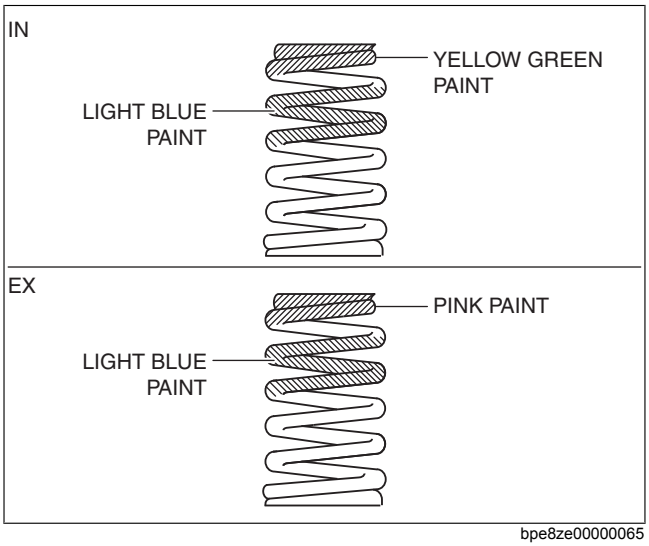
bpe8ze00000020

SKYACTIV-G 2.0 Except MX-5 (ND) VIN Serial No.300001-



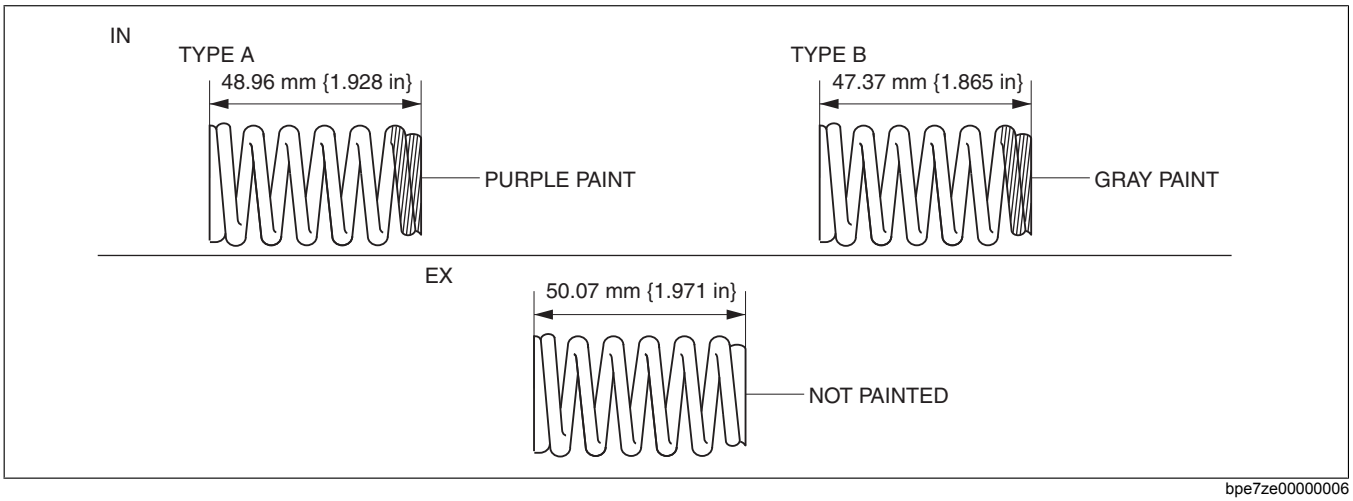
bpe7ze00000005

SKYACTIV-G 2.0 MX-5 (ND) VIN Serial No.300001-



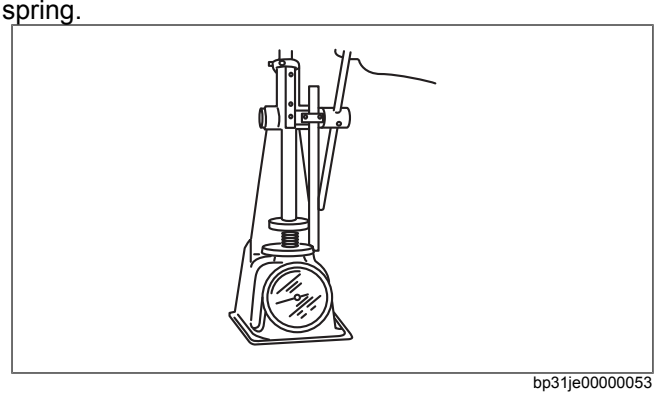
bpe8ze00000065

SKYACTIV-G 2.5



bpe7ze00000006

1. Measure the valve spring height using the spring tester.
   - If it is not within the specification, replace the valve spring.



bp31je00000053

Valve spring installation height [SKYACTIV-G 1.3]
IN: When pressurized with spring force of 161—179 N {16.5—18.2 kgf, 36.2—40.2 lbf}, spring height is 34.0 mm {1.34 in}
EX: When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 34.0 mm {1.34 in}

Valve spring installation height [SKYACTIV-G 1.5] (Except MX-5 (ND))
IN: When pressurized with spring force of 161—179 N {16.5—18.2 kgf, 36.2—40.2 lbf}, spring height is 34.0 mm {1.34 in}
EX: When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 34.0 mm {1.34 in}

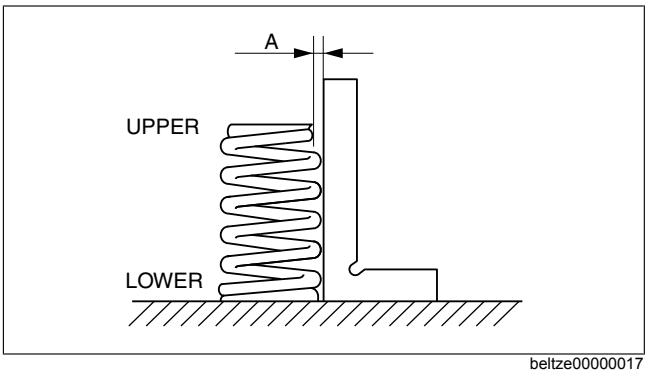
Valve spring installation height [SKYACTIV-G 1.5] (MX-5 (ND))
IN: When pressurized with spring force of 161—179 N {16.5—18.2 kgf, 36.2—40.2 lbf}, spring height is 34.0 mm {1.34 in}
EX: When pressurized with spring force of 228—252 N {23.3—25.6 kgf, 51.3—56.6 lbf}, spring height is 34.0 mm {1.34 in}

Valve spring installation height [SKYACTIV-G 2.0]
IN: When pressurized with spring force of 190—210 N {19.4—21.4 kgf, 42.8—47.2 lbf}, spring height is 38.0 mm {1.50 in}
EX: When pressurized with spring force of 228—252 N {23.3—25.6 kgf, 51.3—56.6 lbf}, spring height is 38.0 mm {1.50 in}

Valve spring installation height [SKYACTIV-G 2.5, type A]
When pressurized with spring force of 228—252 N {23.3—25.6 kgf, 51.3—56.6 lbf}, spring height is 38.0 mm {1.50 in}

Valve spring installation height [SKYACTIV-G 2.5, type B]
IN: When pressurized with spring force of 195—215 N {19.9—21.9 kgf, 43.9—48.3 lbf}, spring height is 38.0 mm {1.50 in}
EX: When pressurized with spring force of 228—252 N {23.3—25.6 kgf, 51.3—56.6 lbf}, spring height is 38.0 mm {1.50 in}

2. Measure the amount of off-square on the valve spring using a square.
   (1) Rotate the valve spring one full turn and measure A at the point where the gap is the largest.
      • If it exceeds the maximum specification, replace the valve spring.



beltze00000017

Maximum valve spring off-square [SKYACTIV-G 1.3, SKYACTIV-G 1.5]
IN: 2.0 ° (3.3 mm {0.1299 in})
EX: 2.0 ° (3.4 mm {0.1339 in})

Maximum valve spring off-square [SKYACTIV-G 2.0]
IN: 2.0 ° (3.5 mm {0.1378 in})
EX: 2.0 ° (3.6 mm {0.1417 in})

Maximum valve spring off-square [SKYACTIV-G 2.5]
IN: 2.0 ° (3.6 mm {0.1417 in})
EX: 2.0 ° (3.6 mm {0.1417 in})